

VRDots — MC and Db Swap Conditions: All MCS Sessions

CUED = delayed-onset field translates | UNCUED = always-visible field translates | Db label flipped in analysis to restore onset-cue convention | Accuracy: response within ±67.5° of TransDeg | 2026-04-30

Asset Parameters (verified from YAML files)		Ap r (°)	N / field	Dens /deg²	Dur (ms)	Conditions run	fixScale Factor	Excl r (°)	Fix arm eff (°)
DenseCM	DensityCompare_Peak_ColorMotionSwap_v1	3.5	500	13.0	80	N, MC	1.0 ‡	1.1	1.000 ‡
S&B Replication	StonerBlanc_Replication_v1	2.0	63	5.0	44	N, MC	0.222	0.396	0.088
MCvsDb Ap1.65	SubfieldSwap_MCvsDb_Ap165_v1	1.65	43	5.0	80	MC, Db (+N pooled from CatekExact)	0.47	0.5	0.235
CatekExact Ap1.65	SubfieldSwap_CatekExact_v1	1.65	43	5.0	80	N, D, Da, Db	0.47	0.5	0.235
CatekExact NDb	SubfieldSwap_CatekExact_NDb_v1	1.65	43	5.0	80	N, Db	0.47	0.5	0.235
HighDens Ap3.5	SubfieldSwap_HighDens_v1	3.5	173	4.5	80	N, D, Da, Db	1.0 ‡	1.1	1.000 ‡
Catek Ap3.5	SubfieldSwap_Catek_v1	3.5	43	1.1	80	N, D, Da (Db absent – session ended early)	1.0 ‡	1.1	1.000 ‡

‡ fixationScaleFactor absent from YAML → C# default 1.0; Fix arm eff = fixationCrosshairArmLength_deg × fixationScaleFactor; Exclusion radius NOT scaled.

Results — All MCS Sessions (fixed duration)

Label	Sessions (n total)	Swap	CUED %	UNCUED %	Δ (pp)	Sig	n cued	n uncued	Notes
DenseCM	260423_1053 (n=1024)	N	83.6	46.1	+37.5	***	256	256	fixScale‡ Ap3.5° 500/field 80ms
		MC	69.9	59.8	+10.2	*	256	256	
S&B Replication	260430_1312+1512 (n=11026)	N	65.2	48.6	+16.6	***	256	257	Ap2.0° 63/field 44ms fixScale=0.222
		MC	70.7	50.6	+20.1	***	256	257	
MCvsDb Ap1.65	260429 ×3 (n=768 MC+Db; N=pooled CatekExact n=345)	N	83.4	68.8	+14.6	**	175	170	N pooled from CatekExact (same params)
		MC	84.4	72.9	+11.5	**	192	192	
		Db	81.8	71.4	+10.4	*	192	192	
CatekExact Ap1.65	260427_0707+1003 (n=873)	N	86.5	63.2	+23.3	***	111	106	Db label flipped
		D	87.9	69.1	+18.8	***	107	110	
		Da	65.2	84.1	-18.9	**	112	107	
		Db	84.5	73.6	+10.9	*	110	110	
CatekExact NDb	260427_1217 (n=256)	N	78.1	78.1	+0.0	n.s.	64	64	Db label flipped
		Db	84.4	71.9	+12.5	†	64	64	
HighDens Ap3.5	260427_1423 (n=512)	N	87.5	56.2	+31.2	***	64	64	fixScale‡ Db label flipped
		D	87.5	42.2	+45.3	***	64	64	
		Da	65.6	68.8	-3.1	n.s.	64	64	
		Db	81.2	75.0	+6.2	n.s.	64	64	
Catek Ap3.5	260424_1801 (n=391)	N	78.1	54.7	+23.4	**	64	64	fixScale‡ session ended before Db
		D	81.2	43.9	+37.3	***	64	66	
		Da	77.3	74.6	+2.6	n.s.	66	67	

Row color = swap type

N

MC

Db

D

Da

DenseCM
Ap3.50p, 500/field, 80ms

VR Dotis Cue

Effect (top) and

Absolute Accuracy

Gettysburg NDb
65, 43/field, 80ms (cha

ghDens Ap3.5°
73.7 (field), 80ms

